Statistical Decision Making
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Broadening the Domain of Application of Regression Models:
Scaling Coefficients
Indicator Variables

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Diamonds Varying with Two Characteristics

Our regression of Price against Carat and Clarity from our previous slide deck is shown below.
We estimated the model and viewed the results using the following commands.

```
# Estimate Diamond regression
DiamReg = tprstats.model("cs", "Price~Carat+Clarity", Diamonds_211)
```

```
# view the regression summary
DiamReg.summary()
```

Dep. Variable:	Price	**R-squared:**	0.950
Model:	OLS	**Adj. R-squared:**	0.950
No. Observations:	211	**F-statistic:**	773.8
Covariance Type:	HC1	**Prob (F-statistic):**	4.54e-97

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	-1.16e+04	759.549	-15.278	0.000	-1.31e+04	-1.01e+04
Carat	1.193e+04	314.094	37.973	0.000	1.13e+04	1.25e+04
Clarity	1308.8177	124.664	10.499	0.000	1063.051	1554.584

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RHS Variables – Interpreting Coefficients

Recall that the coefficient, b_j, of a right-hand-side variable tells us the predicted change in y for a one unit change in x_j while holding all other variables constant.

- Our regression tells us that an increase in clarity by one unit would increase predicted price by $1,309 holding carat weight constant.
- Our regression tells us that a one carat increase in weight holding clarity constant would increase predicted price by $11,930.

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Thinking about Magnitude of a Coefficient

You need to consider the scaling of your variables to judge magnitude.

Example: A carat is 200 milligrams. Our coefficient for carat is $11930.

- Suppose we measure weight in milligrams instead of carats.
  - What would the coefficient be?
  - The coefficient would be $11930/200=$59.65
  - A one milligram increase in weight would increase predicted price by $59.65
- Suppose we measure weight in grams instead of carats:
  - There are five carats per gram. What would the coefficient of gram be?

Very Important:

- You cannot judge the importance of a coefficient simply by looking at how big the coefficient is.
- You need to judge the coefficient relative to the scaling of the variable.

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Recall The Definition of Elasticity

Elasticity is an important measure used widely in economics and business.

In your managerial economics course, you learned the definition of elasticity.

By way of review: Suppose you have an equation in which y is the dependent variable and x_j is an independent variable.

Definition: The elasticity of y with respect to x_j is the percent by which y changes when x_j changes by one percent.

Be sure you memorize this definition.

Elasticity is a scale-free measure. The elasticity of y with respect to x_j does not depend on the units in which we measure either y or x_j.

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Summary Statistics for Diamond Data

In the previous slide deck, we calculated the summary statistics for the diamond data set using the describe() command.

We will use values from this table in the calculations that follow.

```
# view summary statistics
Diamonds_211.describe()
```

	Price	Carat	Clarity	Color	Cut	Salesperson
count	211.000000	211.000000	211.000000	211.0	211.0	211.000000
mean	8769.364929	1.239573	4.270142	11.0	2.0	5.208531
std	574[illegible]00127	0.511450	1.206336	0.0	0.0	2.469649
min	2056.000000	0.510000	2.000000	11.0	2.0	1.000000
25%	4037.500000	0.800000	3.000000	11.0	2.0	3.000000
50%	6914.000000	1.110000	4.000000	11.0	2.0	5.000000
75%	11750.500000	1.585000	5.000000	11.0	2.0	7.000000
max	28546.000000	2.470000	8.000000	11.0	2.0	9.000000

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Illustration of Calculating Elasticity for A Linear Equation

The way in which an elasticity is calculated depends on the functional form. Consider calculating elasticity for a linear equation.

Because elasticity is not constant along a linear equation, it is common to calculate the elasticity at the sample means. Evaluated at the sample mean, the elasticity of y with respect to x_j is $\frac{\bar{x}_j}{\bar{y}}\frac{dy}{dx_j}$. In a linear regression, $\frac{dy}{dx_j} = b_j$.

Hence, in a linear regression, the elasticity can be expressed as: $b_j \frac{\bar{x}_j}{\bar{y}}$.

Consider the coefficient of Carat in our above regression of Price against Carat and Clarity.

The derivative of Price with respect to Carat is 11930. From the summary statistics, we see that the mean of Price is 8769.4 and the mean of Carat is 1.2396.

Hence, the elasticity is: $b_j \frac{\bar{x}_j}{\bar{y}} = 11930 * \frac{1.240}{8769.4} = 1.69$

Interpretation: A one percent increase in carat increases predicted price by 1.69 percent.

Note: We would obtain the same elasticity if we measure weight in milligrams instead of carats.

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Standardized Coefficients

Another scale-free measure is the Standardized Coefficient.

Suppose you have an equation in which y is the dependent variable and x_j is an independent variable.

Definition: The Standardized Coefficient of x_j is the number of standard deviations by which y changes if x_j changes by one standard deviation.

This measure is less widely used in business than the elasticity, but it is gaining increased usage. There are some applications in which standardized coefficients are used frequently. For example, in research on education, standardized coefficients are used frequently.

The next slides illustrates the calculation of a standardized coefficient.

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Illustration of Calculating A Standardized Coefficient

Let's calculate the standardized coefficient of Clarity in our diamond model.

The coefficient of Clarity is 1309.

The standard deviations of Clarity and Price are 1.206 and 5744.

Calculate the standardized coefficient of Clarity:

- It is the coefficient of Clarity times the standard deviation of Clarity divided by the standard deviation of Price: 1309*1.2063/5744 = .2749

Interpretation: A one standard deviation increase in Clarity predicts a .275 standard deviation increase in Price.

Note: We would obtain the same result if we measured Price in Yen instead of dollars. We would also get the same result if we defined a new clarity measure such as: ClarityNew= Clarity/8.

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RHS Variables – Interpreting Coefficients

The following Python command does the work of calculating scaled coefficients: *scaledCoefficients(DiamReg)*

A 1% increase in carat yields a predicted 1.69% increase in price.

A one standard deviation increase in clarity increases predicted price by .275 standard deviations of price.

	coefs	std_coefs	elasticities
Carat	11927.145390	1.061928	1.6859
Clarity	1308.817663	0.274854	0.6373

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Summary Comments on Scaling Coefficients

We saw that the magnitude of a coefficient depends on the scaling of a variable.

Hence, we cannot say that one variable is more important than another based on magnitudes of coefficients.

We saw two methods of scaling coefficients, standardized coefficients and elasticities.

Elasticities are frequently used in business applications. Be sure you know this concept well.

"Standardized Coefficients" can be very useful to you in assessing relative magnitude of effects of right-hand-side variables.

In education research, use of standardized coefficients is quite common. For example, a researcher might find that a one standard deviation increase in GMAT increases graduating GPA of MBA students by .3 standard deviations.

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Broadening the Domain of Application of Regression
Indicator Variables

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Motivating Examples

Indicator Variables:

There are many situations in which a right-hand side variable takes on a value of either 0 or 1. Some examples:

-Experiments: Each individual in the data is assigned to a treatment group or to a control group. We code a 0 for each person assigned to the control group and 1 for each person assigned to treatment, (or we could code 0 for treatment and 1 for control).

-Before-After Analysis: To study whether accident rates changed when the speed limit was increased on the Pennsylvania Turnpike, we could gather daily data on accident rates over a period before and after the change in the speed limit, coding 0 for days before the change and 1 for days after.

-Different Product Categories: In our study of tire pricing later in the course, different tire types are coded by indicator variables (e.g., 1 if designed for all terrain, 0 otherwise)

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Indicator Variables

An indicator variable takes on values of 0 or 1.

Examples are the following three indicator variables from the water conservation case:

- Treat1 equals 1 if a household was assigned to treatment one and zero otherwise.
- Treat2 equals 1 if a household was assigned to treatment two and zero otherwise.
- Treat3 equals 1 if a household was assigned to treatment three and zero otherwise.

Note that indicator variables are sometimes called "dummy variables" or "fixed effects".

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Indicator Variables for Several Households

-From this list, you can see the treatment or control assignment of each of the first 20 households in our Water Conservation data set.

-A given household can be in only one group.

-A zero in all three columns means that the household was assigned to the control group.

	TREAT1	TREAT2	TREAT3
1	0	0	0
2	0	0	0
3	1	0	0
4	0	0	0
5	0	0	0
6	1	0	0
7	0	0	0
8	0	0	0
9	0	0	1
10	0	0	0
11	0	0	0
12	0	0	0
13	0	0	0
14	0	0	0
15	0	0	0
16	0	0	1
17	0	0	0
18	0	0	0
19	0	1	0
20	0	0	0

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Review Analysis of Treatment 3

Let's review the analysis of Treatment 3 that we did for the water conservation case.

Our data are in file Water_Conservation_Data.xlsx. In the data file:

- Variable GROUP takes on value 1 through 4. Values 1, 2, and 3 denote assignment to treatments 1, 2, and 3, with GROUP 4 being the control group.
- SUMMER_07 is thousands of gallons of water use of each household in the summer of 2007.
- TREAT1 equals 1 for household who received treatment 1 and equals 0 otherwise. TREAT2 and TREAT3 are defined similarly.

We selected households in GROUP 3 or GROUP 4.

```
# Select the subset of observations in treatment group 3 and the control group
WatDat34=Water_Conservation_Data.query('GROUP==3|GROUP==4')
```

We then ran a regression to estimate effect of treatment 3:

```
# Run regression to estimate effect of treatment 3 and view the regression output
WatReg34 = tprstats.model("cs", 'SUMMER_07~TREAT3', WatDat34)
WatReg34.summary()
```

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Estimating Effectiveness of Treatment 3

We found mean water use in the control group was 36,473 gallons

Difference between treatment 3 and control group was -1,614 gallons

We found the treatment effect significant at the 5% level and the 1% level

Dep. Variable:	SUMMER_07	**R-squared:**	0.000
Model:	OLS	**Adj. R-squared:**	0.000
No. Observations:	83319	**F-statistic:**	36.54
Covariance Type:	HC1	**Prob (F-statistic):**	1.50e-09

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	36.4725	0.109	333.730	0.000	36.258	36.687
TREAT3	-1.6145	0.267	-6.045	0.000	-2.138	-1.091

Notes:
[1] Standard Errors are heteroscedasticity robust (HC1)

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Estimating All Treatment Effects At Once

Estimate and view the results of the following regression using the full water conservation dataset:

```
# Estimate all treatment effects at once and view the results.
WatRegAll_07=tprstats.model("cs", "SUMMER_07~TREAT1+TREAT2+TREAT3", Water_Conservation_Data)
WatRegAll_07.summary()
```

This provides the mean of control group and the effect of each treatment.

For example, we estimate that Treatment 2 reduced usage by 1,045 gallons relative to the control group.

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	36.4725	0.109	333.728	0.000	36.258	36.687
TREAT1	-0.1197	0.302	-0.396	0.692	-0.712	0.472
TREAT2	-1.0451	0.282	-3.701	0.000	-1.599	-0.492
TREAT3	-1.6145	0.267	-6.045	0.000	-2.138	-1.091

Note that we have four groups but we only include indicator variables for three of the groups. The intercept captures the fourth group—in this case the control group. We can think of the control group as the **Reference Category** for this regression.

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Check Your Understanding

To confirm the interpretation of b_2, for the regression on previous slide, let's write out the details.

Our estimated model is: $\hat{y} = b_0 + b_1\text{Treat1} + b_2\text{Treat2} + b_3\text{Treat3}$

Consider a household in treatment group 2.

What are the values of the "Treat" variables for this household?

Answer: Treat1=0, Treat2=1, Treat3=0

For this household: $\hat{y} = b_0 + b_1*0 + b_2*1 + b_3*0 = b_0 + b_2 = 36.4725 - 1.0451 = 35.4274$

This is the mean for treatment group 2.

Consider a household in the control group.

What are the values of the "Treat" variables for this household?

Answer Treat1=0, Treat2=0, Treat3=0

For this household: $\hat{y} = b_0 + b_1*0 + b_2*0 + b_3*0 = b_0 = 36.4725$

What is the difference between the above households: $b_0 + b_2 - b_0 = b_2 = 35.4274 - 36.4725 = -1.0451$

This shows that the coefficient of Treat2 on the previous slide (-1.045) is the difference in means between treatment group 2 and the control group. This is the treatment effect of Treatment 2.

The p-value in the output tells us that the effect of Treatment 2 is statistically significant at the 5% level.

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Multiple Regression
Joint Hypothesis Test

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Joint Hypothesis Tests

Your boss is a pessimist. Your boss says:

- You've shown me that the estimated effects of the treatments are different, but I know that these estimates are affected by randomness? Is there some kind of statistical test that we can use to show that the true effects of the treatments aren't all the same?

How can we interpret this question in terms of our model?

- Your boss' null hypothesis is that the population coefficients of all three treatment variables are the same.
- To formulate this as a hypothesis test, write our model as follows:

$$\text{SUMMER_07} = \beta_0 + \beta_1\text{TREAT1} + \beta_2\text{TREAT2} + \beta_3\text{TREAT3} + \varepsilon$$

- Your boss' null hypothesis is the that: $\beta_1 = \beta_2 = \beta_3$
- We can rewrite this as: $\beta_1 = \beta_2$ and $\beta_2 = \beta_3$
- Why? If $\beta_1 = \beta_2$ and $\beta_2 = \beta_3$, then it must be the case that $\beta_1 = \beta_3$
- Using Python, we can test this using the following command:

```
# Test null hypothesis that all populationtreatment effects are the same.
WatRegAll_07.wald_test("TREAT1=TREAT2, TREAT2=TREAT3")
```

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Regression Makes Multiple Tests Easy

- We obtain the result below:
- We find a p-value=.0003. Hence, we strongly reject the hypothesis that all three treatments had the same effect.
- You can tell your boss that the odds are less that one in a thousand of getting by chance such large differences in estimated treatment effects .
- This application shows the value of the wald_test() command for testing multiple coefficients at one time.

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Automated Creation of Indicator Variables

Our data file contains the indicator variables TREAT1, TREAT2, and TREAT3. This makes it easy for us to to estimate the model that we studied in these notes.

However, in most practical settings, you would not have such indicator variables created for you. Instead, you would have a variable like the GROUP variable. You would then need to create your own indicator variables for the individual treatments.

Fortunately, Python provides an easy way to automate creation of indicator variables when you need them. This is done using the C() command as shown on the next slide

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Using The Factor Command

Issue the following commands:

```
# Automated Creation of Indicator variables: The Category command
WatRegAll_07=tprstats.model("cs", "SUMMER_07~C(GROUP)", Water_Conservation_Data)
WatRegAll_07.summary()
```

You obtain the following result. Notice that the C() command has made GROUP 1 the reference category. We want the control group, GROUP4, to be the reference category. We see how to do that on the next slide.

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	36.3528	0.282	129.105	0.000	35.801	36.905
C(GROUP)[T.2]	-0.9254	0.384	-2.413	0.016	-1.677	-0.174
C(GROUP)[T.3]	-1.4948	0.372	-4.014	0.000	-2.225	-0.765
C(GROUP)[T.4]	0.1197	0.302	0.396	0.692	-0.472	0.712

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Specifying Reference Category with the C() Command

We specify the reference category using the Treatment() option in the C() command. This makes group 4 the reference category.

```
WatRegAll_07=tprstats.model("cs", "SUMMER_07~C(GROUP, Treatment(4))", Water_Conservation_Data)
WatRegAll_07.summary()
```

We obtain the result below.

Scroll back and you will see that the result below is the same as we obtained with TREAT1, TREAT2, and TREAT3.

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	36.4725	0.109	333.728	0.000	36.258	36.687
C(GROUP, Treatment(4))[T.1]	-0.1197	0.302	-0.396	0.692	-0.712	0.472
C(GROUP, Treatment(4))[T.2]	-1.0451	0.282	-3.701	0.000	-1.599	-0.492
C(GROUP, Treatment(4))[T.3]	-1.6145	0.267	-6.045	0.000	-2.138	-1.091

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Broadening the Domain of Application of Regression Models:
Regression Models with Both Indicator Variables and Continuous Variables

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Models with Continuous and Indicator Variables

The goal of the next application we study is to help you gain an understanding of models with both continuous variables and indicator variables.

In design of delivery routes, it is useful to estimate models that capture the variation in time required to deliver on different routes.

We illustrate with a parcel delivery application with three delivery routes.

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Combining Continuous and Indicator Variables Cont'd

Parcel Delivery Application

- *On Time Delivery* (OTD) is a parcel delivery service.
- Each day, each driver goes to the OTD depot, drives to the delivery route, delivers the parcels, and returns the van to the depot.
- OTD is developing a regression model to benchmark delivery times for its routes.
- OTD has collected daily data on three delivery routes, denoted A, B, and C.
- OTD has obtained data for 31 days for route A, 27 for route B, and 42 for route C.
- Variables are PARCELS delivered each day and total delivery time in minutes (DELTIM) from the time the driver leaves the depot to the time the driver returns to the terminal.
- OTD has created **indicator variables**. If an observation is for a delivery on route A, variable RA is set equal to one, otherwise RA=0. Variables RB and RC are defined similarly.

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Indicator and Continuous Variables

A few daily observations are listed below.

To be sure you understand the interpretation, please be sure you can answer for each observation:

- Which route?
- How many parcels that day on that route?
- How many minutes of delivery time?

RA	RB	RC	PARCELS	DELTIM
0.000000	1.000000	0.000000	29.00000	219.0000
0.000000	0.000000	1.000000	24.00000	226.0000
0.000000	0.000000	1.000000	32.00000	271.0000
0.000000	1.000000	0.000000	46.00000	394.0000
0.000000	1.000000	0.000000	32.00000	267.0000
1.000000	0.000000	0.000000	59.00000	495.0000

The first observation is a delivery on Route B of 29 parcels in 219 minutes.

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Modeling Delivery Time

```
# Import Parcel Delivery Data Set
Parcel_Delivery = pandas.read_excel("Parcel_Delivery.xlsx")
```

OTD has estimated the following regression

```
# Parcel Delivery Regression
ParcelReg=tprstats.model("cs", "DELTIM~PARCELS+RB+RC", Parcel_Delivery)
ParcelReg.summary()
```

In the following slides, we will interpret each coefficient.

OLS Regression Results

Dep. Variable:	DELTIM	**R-squared:**	0.976
Model:	OLS	**Adj. R-squared:**	0.975
No. Observations:	100	**F-statistic:**	1488.
Covariance Type:	HC1	**Prob (F-statistic):**	2.58e-80

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	17.6083	5.767	3.053	0.003	6.160	29.057
PARCELS	7.9919	0.120	66.479	0.000	7.753	8.231
RB	9.2761	4.973	1.865	0.065	-0.595	19.148
RC	13.3647	4.877	2.740	0.007	3.683	23.046

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Interpretation

In models with both continuous and indicator variables, the indicator variables permit the intercepts to differ by category.

- In this application, the intercepts are estimates of fixed costs.
- The fixed costs (minutes) for routes A, B, and C are 17.61, 26.88, and 30.97.

The estimated time per additional parcel (slope) is 7.99 minutes; this regression assumes the slope to be the same for all three routes. We will generalize later.

Note that we have not included RA in the regression.

- This makes route A the reference category. [The reference category is sometimes called the "excluded" category, but it is important to understand that data for all routes are included in the regression. No data are excluded.]
- The coefficient of RB is the estimated **difference in fixed time between routes A and B** (9.27 minutes).
- The coefficient of RC is the estimated **difference in fixed time between routes A and C** (13.36 minutes). Yes, really, between A and C, not B and C.

Be sure you see the relationship between the regression on the previous page and the figure on the next page. It is absolutely essential to understanding models with both continuous and indicator variables.

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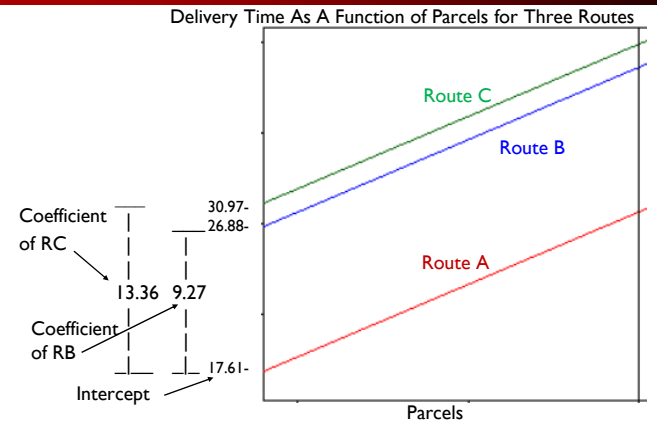
Illustration of Delivery Time Model



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A Brief Digression

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Automatic Creation of Indicators for This Model

We saw earlier that *Python* automates the process of creating indicator variables (factors) when you have a single variable denoting all of the indicator categories.

- In our Parcel_Delivery file, ROUTE_NAME gives the name of each route.
- This kind of coding of routes in a single column would be quite natural in practice.
- Estimate and view the results using:

```
# Parcel Delivery Regression with factors based on ROUTE_NAME
ParcelRegFac= tprstats.model("cs", 'DELTIM~PARCELS+C(ROUTE_NAME)', Parcel_Delivery)
ParcelRegFac.summary()
```

The C() command tells Python to make a separate indicator variable for each distinct item in the variable ROUTE_NAME.

This is particularly convenient in situations where you have a large number of factors (e.g., a large number of routes).

Python reports the results with the indicator variables ordered alphabetically. The first alphabetically is made the reference category (RA in our current application) .

Our regression result is on next slide.

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Result

Scroll back and you will see the results below are the same as we obtained previously when we used a separate column for each route.

OLS Regression Results

Dep. Variable:	DELTIM	**R-squared:**	0.976
Model:	OLS	**Adj. R-squared:**	0.975
No. Observations:	100	**F-statistic:**	1488.
Covariance Type:	HC1	**Prob (F-statistic):**	2.58e-80

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	17.6083	5.767	3.053	0.003	6.160	29.057
PARCELS	7.9919	0.120	66.479	0.000	7.753	8.231
C(ROUTE_NAME)[T.RB]	9.2761	4.973	1.865	0.065	-0.595	19.148
C(ROUTE_NAME)[T.RC]	13.3647	4.877	2.740	0.007	3.683	23.046

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Factors For Numbered Routes

If the categories in your data are designated by numbers instead of names, the C() command can be used as before.

The ROUTE_NUMBER variable in the data file denotes the routes by numbers 1, 2, and 3.

Estimate and view the results using:

```
# Parcel Delivery with factors based on ROUTE_NUMBER
ParcelRegFac= tprstats.model("cs", 'DELTIM~PARCELS+C(ROUTE_NUMBER)', Parcel_Delivery)
ParcelRegFac.summary()
```

We obtain the same estimates as on the previous slide:

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	17.6083	5.767	3.053	0.003	6.160	29.057
PARCELS	7.9919	0.120	66.479	0.000	7.753	8.231
C(ROUTE_NUMBER)[T.2]	9.2761	4.973	1.865	0.065	-0.595	19.148
C(ROUTE_NUMBER)[T.3]	13.3647	4.877	2.740	0.007	3.683	23.046

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Broadening the Domain of Application of Regression Models:
Controlling for Pre-Treatment Variables in an Experiment

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Combining Continuous and Indicator Variables

Continuous and indicator variables are frequently used together in a regression model, as in the Parcel Delivery model.

Another application is presented in the following slides.

This illustration makes a very important point about evaluating experiments.

- Controlling for pre-treatment variables can improve precision of estimated treatment effects.
- We will see that doing so establishes that Treatment 3 did in fact have a significant effect in 2009!

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Including Variables Determined Prior to An Experiment

If variables determine prior to an experiment are available, including such variables in our regressions can improve the accuracy of our estimation of the treatment effect.

- Our water conservation case provides a nice illustration
- Water use in 2006 and in April-May of 2007 were determined before the assignment of households to treatment and control groups.
- Let's include those variables in a regression for Summer 2009 regression along with the treatment indicator variables.

The slides that follow compare the results of the following two regressions

```
# Summer 2009 with only treatment variables
Wat2009Reg1=tprstats.model("cs", "SUMMER_09~TREAT1+TREAT2+TREAT3", Water_Conservation_Data)
Wat2009Reg1.summary()
```

```
# Summer 2009 with treatment variables and variables determined prior to the experiment
Wat2009Reg2=tprstats.model("cs", "SUMMER_09~TREAT1+TREAT2+TREAT3+WATER_2006+APR_MAY_07", Water_Conservation_Data)
Wat2009Reg2.summary()
```

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Does Effect of Treatment Persist in 2009?

Recall that the experiment was conducted in Summer of 2007

The case write-up says treatment 3 had a significant effect in Summer 2009,

but we do not find a significant effect effect for TREAT3 in the regression below.

OLS Regression Results

Dep. Variable:	SUMMER_09	**R-squared:**	0.000
Model:	OLS	**Adj. R-squared:**	0.000
No. Observations:	106669	**F-statistic:**	1.672
Covariance Type:	HC1	**Prob (F-statistic):**	0.171

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	27.4558	0.078	350.668	0.000	27.302	27.609
TREAT1	0.3005	0.210	1.433	0.152	-0.110	0.711
TREAT2	-0.0865	0.210	-0.412	0.680	-0.498	0.325
TREAT3	-0.2838	0.197	-1.441	0.150	-0.670	0.102

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Controlling for Pre-Treatment Variables

Because treatments are randomly assigned, we obtain **unbiased coefficient estimates** without controlling for any other variables. Thus, the estimates reported on the previous slides are unbiased.

But, controlling for variables observed prior to treatment can increase **precision** of our estimates.

- WHY?

There are two causes of variation in water use:

- Differences due to experimental treatment.
- Differences across households (e.g., as regflected in differences in water use before the experiment).

The more we can control for variation due to differences households' propensity to use water, the better we can isolate the variation due to the experiment.

- Controlling for pre-treatment variables gives us **more precise estimates** of the treatment effects of our experiment.

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Controlling for Pre-Treatment Variables for Summer 2009

We include water use in 2006 and April-May 2007. Both were determined before the experiment.

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	9.7619	0.558	17.489	0.000	8.668	10.856
TREAT1	0.2385	0.170	1.402	0.161	-0.095	0.572
TREAT2	-0.0587	0.169	-0.348	0.728	-0.389	0.272
TREAT3	-0.3492	0.163	-2.139	0.032	-0.669	-0.029
WATER_2006	0.1850	0.010	18.223	0.000	0.165	0.205
APR_MAY_07	0.4349	0.024	17.827	0.000	0.387	0.483

Compared to the previous regression, the coefficient of treatment 3 is now significant at the 5% level in summer 2009.

Average water use for treatment 3 in 2009 was 349 gallons lower than for the control group.

Notice treatments 1 and 2 are not significant. This reinforces the conclusion that those treatments did not have lasting effects.

The above exactly replicates column (C) of Table 1 of the case writeup.

Key Message: When variables observed **prior** to treatment are available, it is beneficial to include them in the regression because including them gives more precise estimates of the treatment effects.
A more precise estimate may not show a more significant treatment effect.
Increased accuracy might show effects to be smaller (e.g Treatments 1 & 2).

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Were Assignments to Treatment Groups Really Random?

We have data on water usage by each household in year 2006, the year before assignments to treatment were made.

If assignments of households to treatment groups were random, there should be no significant difference in water use by groups in the year before the treatment assignments were made. We test this with the following regression.

```
WatReg06=tprstats.model("cs","WATER_2006~C(GROUP)",Water_Conservation_Data)
WatReg06.summary()
```

We find that there was no significant difference in water usage across groups in 2006.
This supports the claim that assignment to treatment and control groups was indeed random.

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	58.4265	0.370	157.990	0.000	57.702	59.151
C(GROUP)[T.2]	-0.2509	0.531	-0.472	0.637	-1.293	0.791
C(GROUP)[T.3]	0.0091	0.528	0.017	0.986	-1.025	1.043
C(GROUP)[T.4]	-0.1283	0.401	-0.320	0.749	-0.914	0.657

This is called a Balance Check.

Summary

We studied scaling of coefficients.

We defined the concept of an indicator variable.

- We noted that indicator variables are also sometimes called dummy variables or fixed effects.
- We saw how to use the C() command to automate creation of indicator variables.

We estimated models with both Indicator and Continuous Variables

We showed how to use multiple regression to estimate multiple treatment effects in a single equation.

We showed how to use the wald_test() to test the null hypothesis that all three treatment effects are the same.

- As noted in a previous slide deck, the test for multiple coefficients is named after Abraham Wald who developed the test.